Cyclistic Bike Share Case Study

Introduction

The Cyclistic Bike Share Case Study is a capstone project for the Google Data Analytics Professional Certificate on Coursera. In this project, I will follow the data analysis process which I learned from the course: ask, prepare, process, analyze, share and act to analyze the data.

Background

Cyclistic is a bike-share company in Chicago that launched its program in 2016. Since then, it has expanded to a fleet of 5,824 bikes linked to 692 stations across the city. Riders can unlock a bike at one station and return it to another at any time. To meet diverse customer needs, Cyclistic offers single-ride passes, full-day passes, and annual memberships. Riders using single-ride or day passes are considered casual riders, while those with annual passes are members. Financial analysis shows that members are much more profitable than casual riders, making them a key driver of long-term growth.

Until now, Cyclistic's marketing strategy has focused on raising awareness and appealing to broad audiences. The flexibility of its pricing plans has helped attract a wide customer base. However, the company recognizes that casual riders already know the service and rely on it for their mobility needs, which presents an opportunity to increase membership conversions rather than solely targeting new users.

The director of marketing believes the company's future success depends on maximizing the number of annual memberships. To achieve this, the team needs to understand how casual riders and members use Cyclistic bikes differently, why casual riders might choose to upgrade, and how digital media can support targeted campaigns. By analyzing historical trip data, the goal is to design an effective marketing strategy that converts more casual riders into loyal members.

Steps

1. Ask

Statement of the business task:

Cyclistic's marketing director has identified annual memberships as the key driver of long-term growth. The business task is to analyse how casual riders, and annual members use Cyclistic bikes differently. **The purpose of this analysis is to generate actionable insights that can guide marketing strategies to convert more casual riders into annual members**, thereby increasing customer retention and long-term revenue.

Three questions will guide the future marketing program:

1. How do annual members and casual riders use Cyclistic bikes differently?

2. Why would casual riders buy Cyclistic annual memberships?

3. How can Cyclistic use digital media to influence casual riders to become members?

2. Prepare

The dataset used in this analysis was provided as part of the Google Data Analytics Certificate Program on Coursera. It contains historical trip data from **Divvy**, a bike-share system in Chicago operated by Lyft. The datasets are publicly available on AWS at https://divvy-tripdata.s3.amazonaws.com/index.html.

The data is organized in CSV files, each representing one month of trip records. Each file includes fields such as ride ID, rideable type, start and end times, station names and IDs, latitude, longitude and rider type.

The data is considered credible, as it comes directly from the official Divvy system, but it has been made available for educational use only.

Licensing and privacy requirements are addressed by anonymizing personally identifiable information (no customer names, contact details, or payment data are included).

To ensure data integrity, I verified that the files could be read without errors, checked for consistent formatting, and confirmed that variable names were standardized across files.

This dataset supports the business task by allowing analysis of rider behaviour patterns, comparing usage between casual riders and annual members.

However, it has some limitations: It does not include demographic information (e.g., age, gender, income). It only captures trip behaviour, so additional external data (such as weather or events) might be needed for deeper insights.

3. Process

The basis for this analysis is 2024 data and the steps for processing the data are as follow:

- Data Combination
- Data Exploration
- Data Cleaning
- Data Analysis

Data Combination

The 12 tables from January 2024 to December 2024 were stacked and combined into a single table. The table consists of 5,860,568 rows.

Data Exploration

As part of the initial data exploration for the 2024 Cyclistic dataset, I conducted a series of SQL queries to assess data quality, structure, and key attributes. The following steps were performed:

- Checked for missing values, anomalies, and data structure.
- Key observations:
    - Some station-related fields (start_station_name/start_station_id, end_station_name/end_station_id) are missing together, so they **cannot be reliably imputed**.
    - ride_duration had a small number of rides with durations <1 minute or >24 hours.
    - rideable_type has three categories: electric, classic, electric_scooter.
- Core columns for analysis: ride_id, rideable_type, started_at, ended_at, member_casua, duration.

Data Cleaning

- Created new columns:
    - ride_duration_mins to store ride duration in minutes.
    - outliers to mark rides with durations <1 minute or >24 hours.
- Removed duplicates:
    - Identified and deleted 211 duplicate rides based on ride_id.
- Standardized data formats:
    - Trimmed and removed carriage returns (\r) from member_casual.
- Removed temporary columns:
    - Dropped row_num used for duplicate identification.
- Final dataset:
    - 5,860,357 rows, ready for analysis.

4. Analyzing

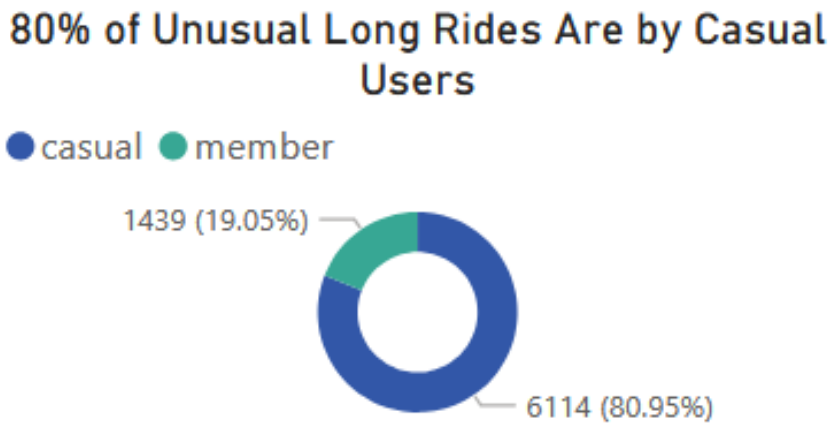
The analysis question is:

How do annual members and casual riders use Cyclistic bikes differently?

The cleaned data is imported into Power BI for analysis and the figures plotted are displayed in the following.

- 80% of Unusual Long Rides Are Taken by Casual Users

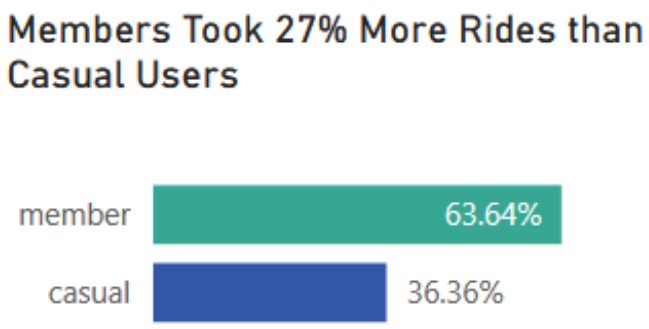
The chart below illustrates the distribution of unusual long rides in 2024 between members and casual riders:

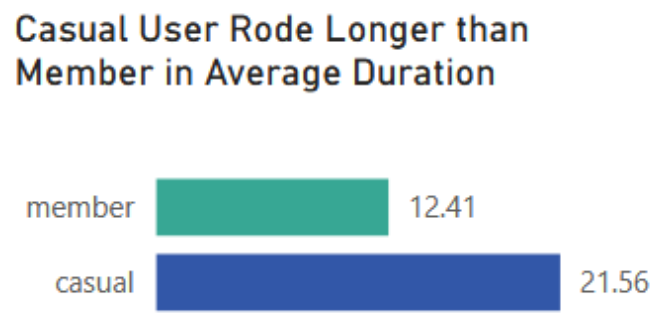


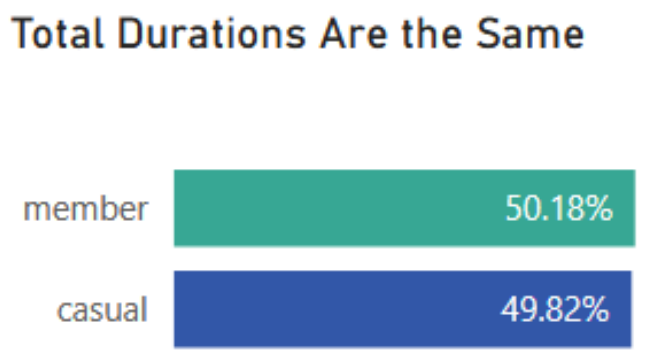
- Casual riders account for **80%** of these rides.
- **Recommendation:** Consider reinforcing the bike rental process to reduce unusually long rides among casual users.

- User Ride Patterns in 2024

The analysis of 2024 ride data shows that **members and casual users contribute similarly to total ride durations**, despite differences in riding patterns:



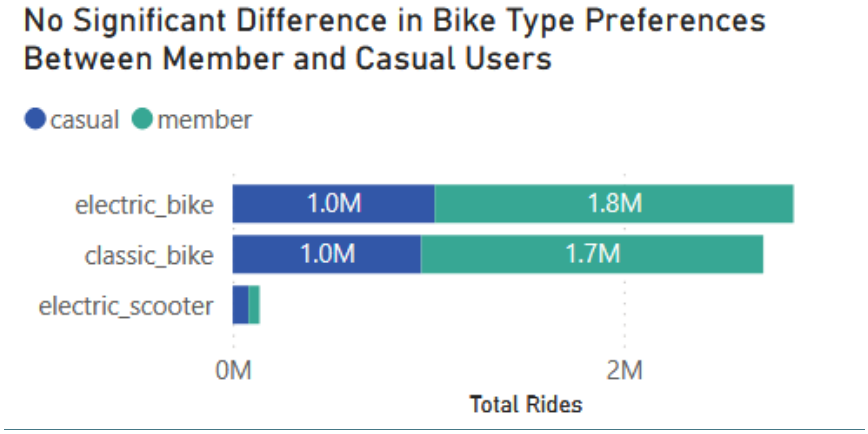




- **Members** take more rides but with shorter average durations.
- **Casual users** ride less frequently but each ride lasts longer.
- **Recommendation:** Implement a member rewards program where members earn points only after achieving both ride-based milestones: completing a minimum number of rides **and** reaching a target cumulative ride duration. Points can be redeemed for small rewards or merchandise. This dual-condition strategy encourages frequent usage, promotes longer engagement per ride, and provides a tangible incentive for casual users to convert to members.

- No Significant Difference in Bike Type Preferences Between Members and Casual Users

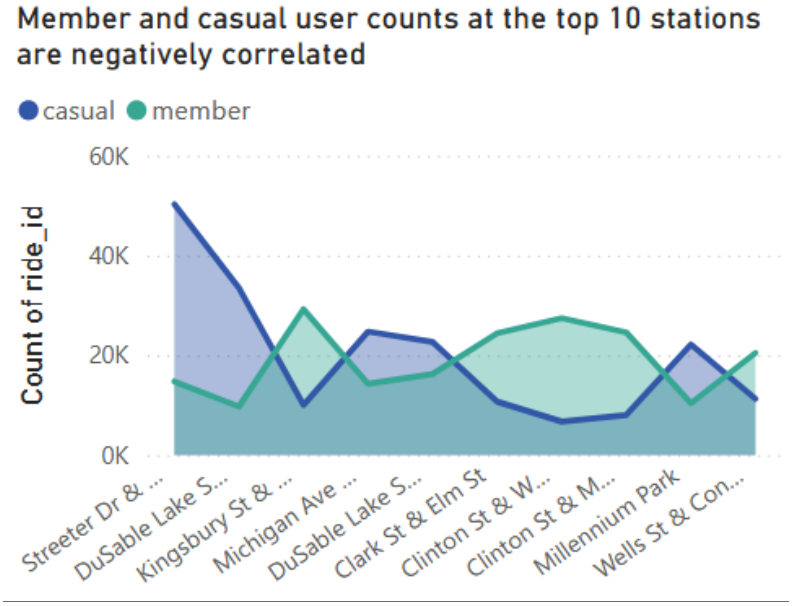
This chart displays the distribution of bike types (electric, classic, electric_scooter) used by members and casual users in 2024.



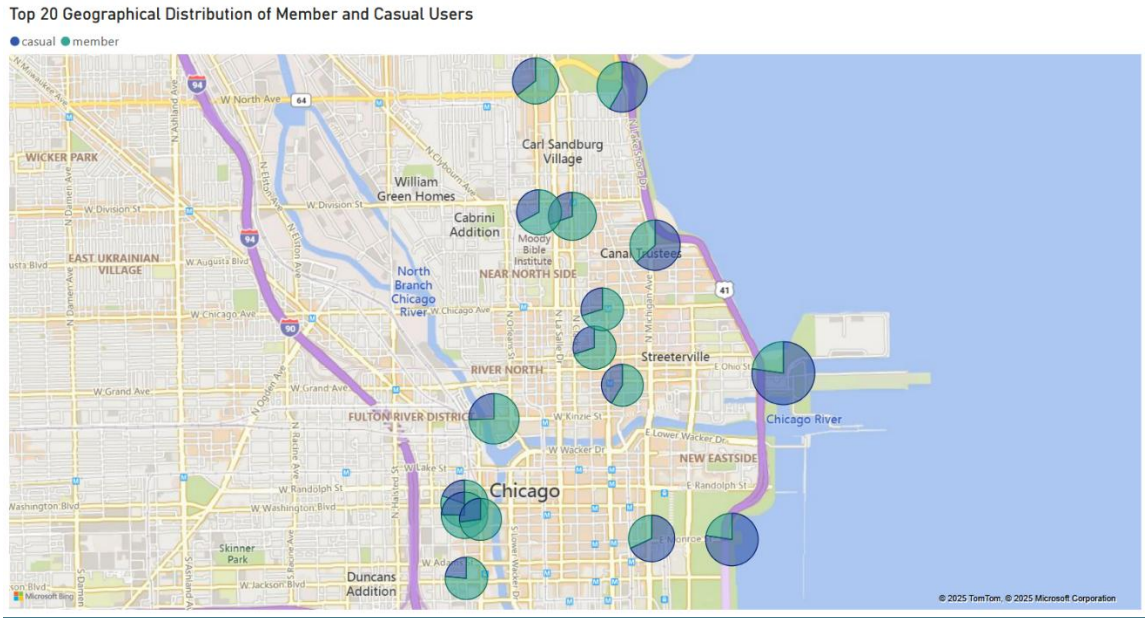
- Members and casual users show **similar preferences** across all bike types.
- Bike type does **not appear to be a key differentiator** between user groups.
- **Recommendation:** No immediate action required, but monitoring bike type usage may help inform **fleet planning** or inventory decisions in the future.

- Member and Casual User Counts at Top Stations Show Distinct Geographic Patterns

This chart displays the number of rides by members and casual users at the top 10 stations in 2024.



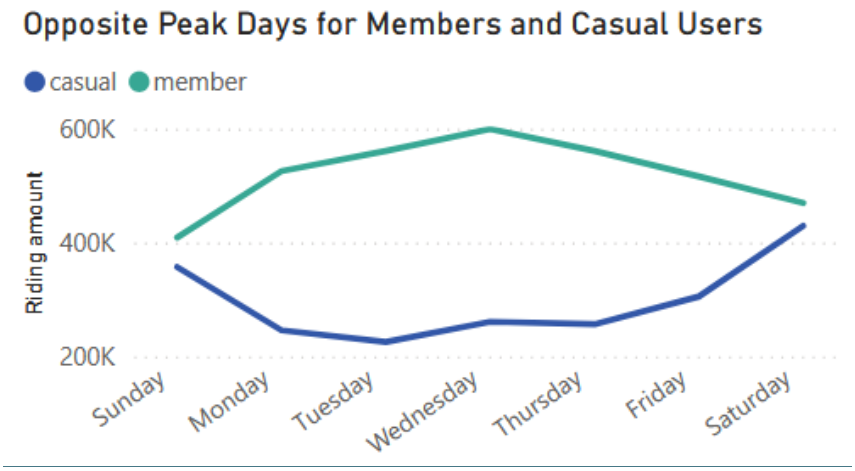
This map displays the top 20 stations by total rides in 2024, showing how member and casual users are distributed across Chicago.

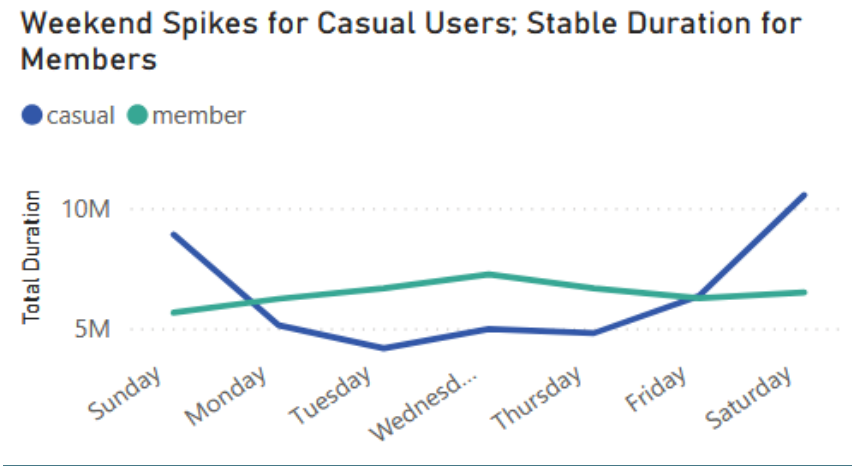


- Member usage is concentrated in downtown and business districts, suggesting weekday commuting patterns.
- Casual user activity is highest near the lakefront, parks, and tourist areas, reflecting leisure and weekend rides.
- **Recommendation:** To increase membership conversion, consider offering a targeted incentive: new members who complete a minimum number of rides in **both urban business areas and recreational/tourist areas** within a month receive a discount (e.g., 10% off) on their following month's membership fee. This dual-region requirement encourages balanced usareas andages high casual-user traffic in recreational areas, and promotes regular engagement across the city.

- Opposite Peak Days for Members and Casual Users

This chart shows the number of rides by members and casual users across days of the week in 2024.

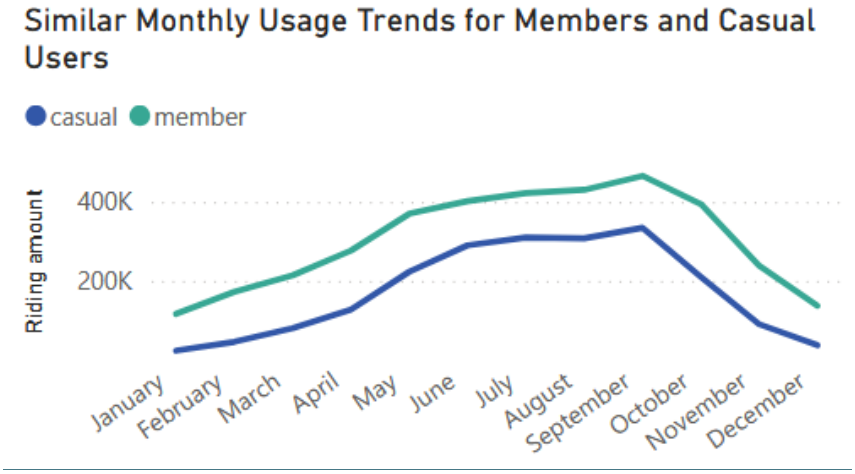


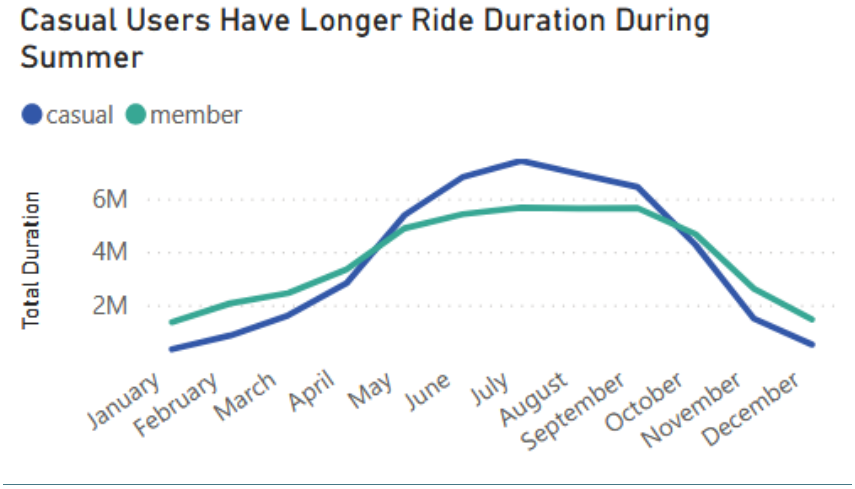


- Members and casual users exhibit opposite weekly behavior patterns.
- Members maintain stable ride duration throughout the week, while casual users show low usage on weekdays and higher usage on weekends.
- **Recommendation:** Implement a time-based pricing strategy, with higher rates on weekends to encourage casual users to convert to members and smooth out demand.

- Similar Monthly Usage Trends for Members and Casual Users, but Casual Users Have Longer Ride Duration During Summer

These charts compare monthly ride counts and average ride duration for members and casual users in 2024.





- Members and casual users show similar monthly usage trends.
- Casual users take longer rides during summer, suggesting leisure or tourist-oriented usage.
- **Recommendation:** Launch a flash winter promotion offering a discount on annual membership fees, e.g., reduced rates for the first three months, to encourage casual users to convert to members during the off-season.

To further investigate the monthly usage trends, I incorporated external data sources including average monthly temperature and public holiday distributions in Chicago for 2024.

The temperature chart reveals a clear seasonal pattern, with peak warmth from June to August aligning closely with high ride volumes, particularly among casual users. This suggests a positive correlation between temperature and ride frequency.

However, September presents an anomaly: casual ride counts remain unusually high despite a noticeable drop in temperature. While the presence of Labor Day (September 2) may partially explain this spike, the month does not contain a significantly higher number of public holidays compared to others.

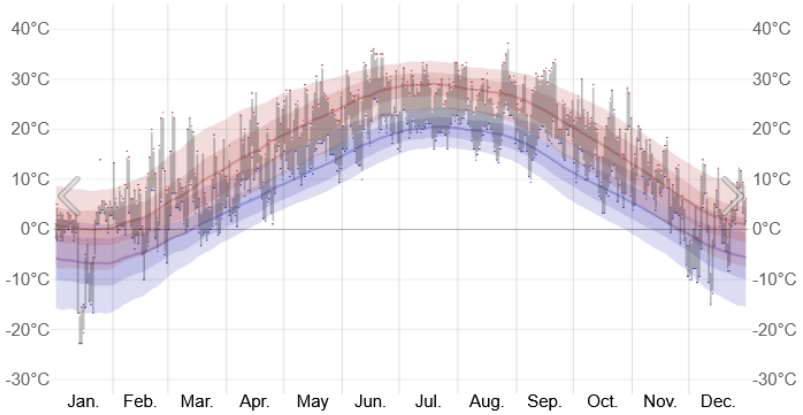
This indicates that additional factors—such as local events, school schedules, or promotional campaigns—may be influencing rider behavior. Further data collection and analysis would be required to fully understand the drivers behind this deviation.

By overlaying these external variables, we gain a more nuanced understanding of seasonal and behavioral patterns, which can inform the timing of promotional campaigns, resource allocation, and membership conversion strategies.

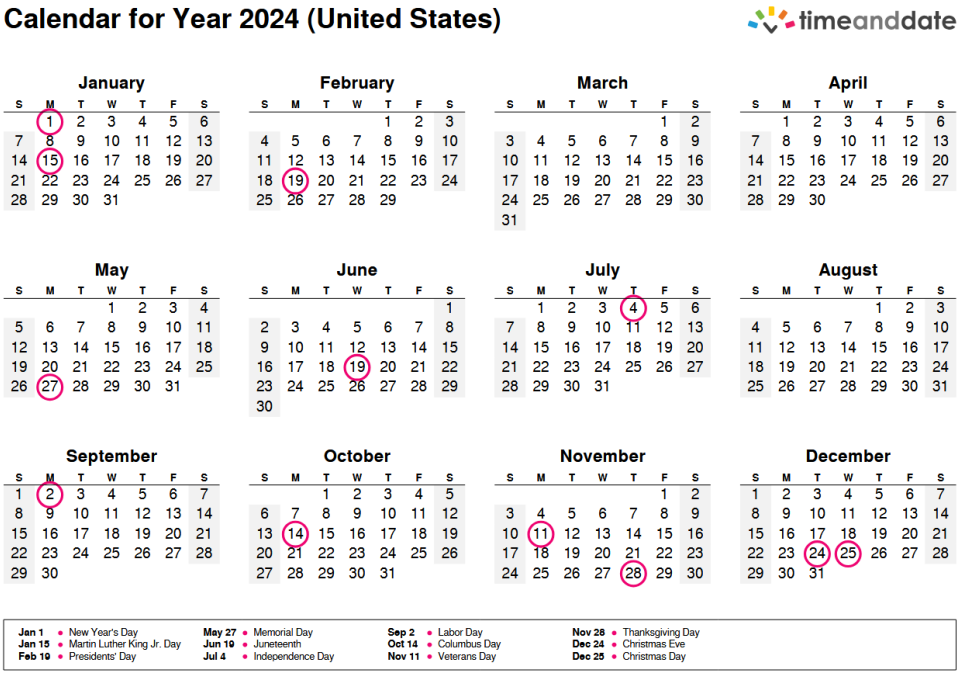
Sources:

- Temperature and precipitation data: *Weatherspark – Historical Weather in Chicago (2024)*
- Public holiday calendar: *Timeanddate.com – U.S. Holidays and Calendar (2024)*

Chicago Temperature History 2024

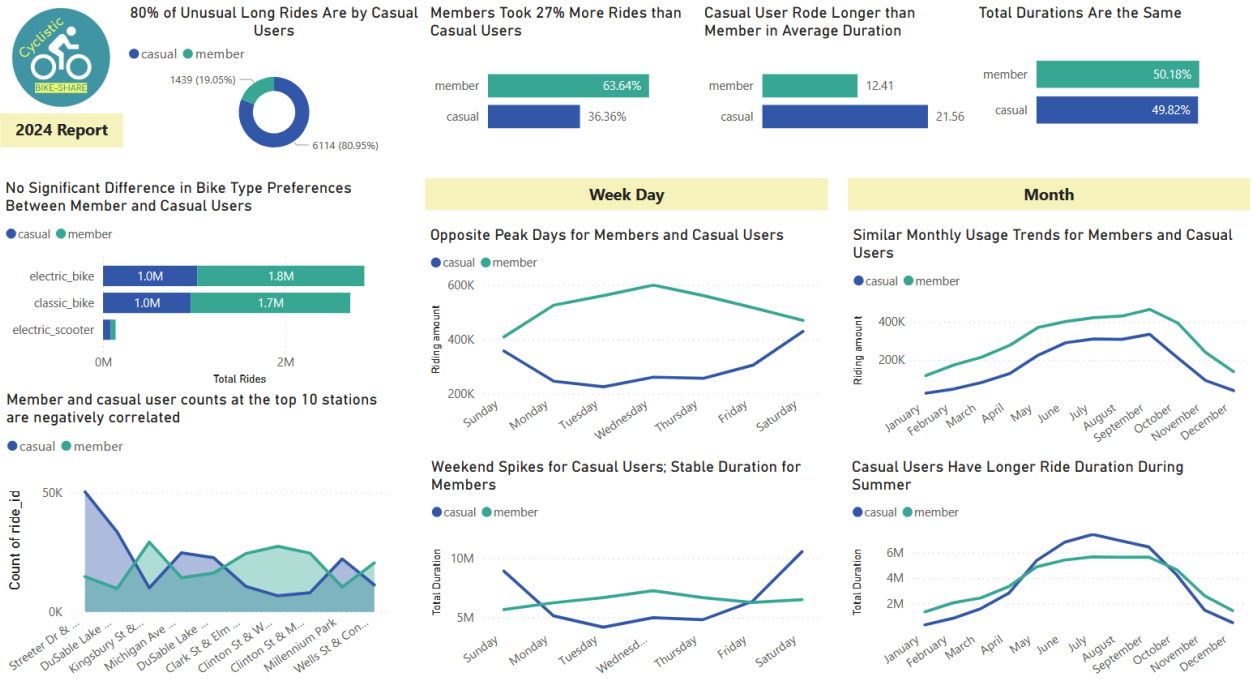


(Sources: https://weatherspark.com/h/y/14091/2024/Historical-Weather-during-2024-in-Chicago-Illinois-United-States)



(Sources: https://www.timeanddate.com/calendar/create.html?hol=9&year=2024)

5.Share



The similiarities and differences between Cyclistic members and casual riders were drawn from the dashboard above.

Similarities:

- Both Cyclistic members and casual riders prefer riding bicycles during the spring and summer seasons (from May to September). However, the number of rides declines after September, likely due to seasonal changes and cooler temperatures that make riding less comfortable.
- Both Cyclistic members and casual riders show a preference for electric bicycles.

Differences:

- Members ride more frequently than casual users.
- Casual users have nearly twice the average ride duration of members (21.56 minutes vs. 12.41 minutes), indicating that casual users mainly ride for leisure, while members tend to use bikes for short trips or commuting.
- Members are more active on weekdays, while casual users peak on weekends, reflecting different usage purposes.
- Usage locations differ: members primarily ride in urban areas, whereas casual users concentrate in coastal, park, or tourist locations.

6.Act– Key Business Decisions to Increase Membership Conversion:

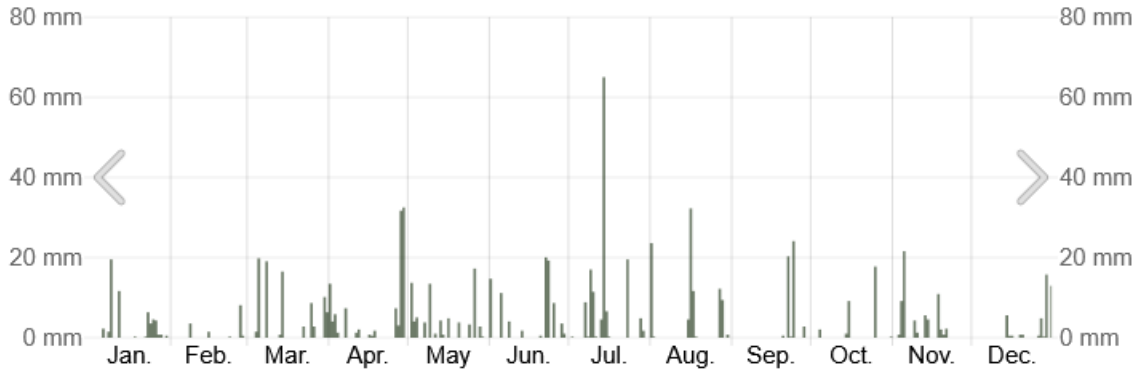
Based on the insights above, I propose the following strategies to improve customer conversion rates and ultimately increase company profit:

1. **Membership Pamphlet:** Promote membership benefits and educate casual users on the advantages of becoming members.
2. **Points Reward System:** Implement a dual-condition rewards program where members earn points only after completing both a minimum number of rides **and** reaching a target cumulative ride duration. Points can be redeemed for small rewards or merchandise.
3. **Targeted Membership Discount:** Offer a discount (e.g., 10% off) to new members who complete a minimum number of rides in both urban business areas and recreational/tourist areas within a month.
4. **Weekend Pricing Strategy:** Increase rates on weekends to encourage casual users to convert to members while smoothing demand.
5. **Winter Flash Promotion:** Introduce a “Warm Up & Ride” winter campaign to boost membership during the off-season.
   Offer double reward points and a small winter-themed gift (e.g., disposable heat pack) for rides completed in winter, or provide a limited-time discounted annual membership (e.g., reduced rates for the first three months) to encourage new sign-ups.

Appendix

Precipitation data was reviewed but did not show a clear correlation with ride frequency. It was therefore excluded from the main analysis.

Daily Precipitation in 2024 in Chicago



(Resource: https://weatherspark.com/h/y/14091/2024/Historical-Weather-during-2024-in-Chicago-Illinois-United-States)